SMART BUS SYSTEM FOR REAL TIME PASSENGER MANAGEMENT SYSTEM

Meet our team

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IoT Based Bus Tracking System And ETA Prediction

Specialization: Information Technology

Project ID: 25-26J-267

Background

- Public transport is the main mode of travel for millions of passengers daily.
- Passengers often face **uncertainty in bus arrival times**.
- If we can predict accurate arrival time of bus, it will save so much time of public tranceport users.

Research Problem

- Existing bus tracking solutions are **not accurate enough**, as they don't consider all real-world conditions..
- **Passenger boarding time** is not included in current ETA predictions
- How to design an IoT-based system that integrates GPS, traffic, weather, and passenger boarding time to provide accurate, real-time ETAs and improve the reliability of public transport?

Research Gap

Features	[1]	[2]	[3]	[4]	Proposed System
GPS-based Tracking	Partial	Partial	✗	✗	✓
Real-Time Traffic Data Integration	Partial	✗	✗	✓	✓
Weather Data Consideration	Partial	✗	✗	✗	✓
Machine Learning for ETA (LSTM/RNN)	Partial	✓	Partial	✓	✓
Passenger Bording time considered	✗	✗	✗	✗	✓
Passenger Mobile/Web Interface	✗	Partial	✗	✗	✓

Research Problem & Solution

Shortcomings of Existing Systems:

- Depend only on GPS and static schedules → inaccurate ETAs.
- Do not integrate **traffic, weather, or Passenger On boarding time**.
- Limited or no **multi-bus tracking**.
- Lack of affordable IoT-based implementations for developing regions.

Proposed Solution:

- IoT devices (GPS + GSM/Wi-Fi) provide **real-time data**.
- **LSTM model** predicts ETAs considering traffic, weather, and contextual factors.
- Supports **multi-bus tracking** with cloud integration.
- **Mobile/Web apps** deliver accurate arrival info to passengers and operators.

Key Domains of Specialization-based Knowledge

- **IoT & Embedded Systems** → Use of Arduino/ESP32, GPS, GSM/Wi-Fi modules for real-time data collection and communication.
- **Cloud Computing & Databases** → Cloud-hosted real-time database (Firebase/AWS DynamoDB) to store and process bus data.
- **Machine Learning & AI** → Application of LSTM (Long Short-Term Memory) for ETA prediction using sequential data.
- **Software Engineering & Web Development** → Mobile and web applications for passengers and administrators to view real-time bus location and ETA.

Latest Technologies Utilized

IoT Hardware

Arduino

GPS (Neo-6M)

GSM(SIM800l)

YL-83 Rain Sensor

Machine Learning

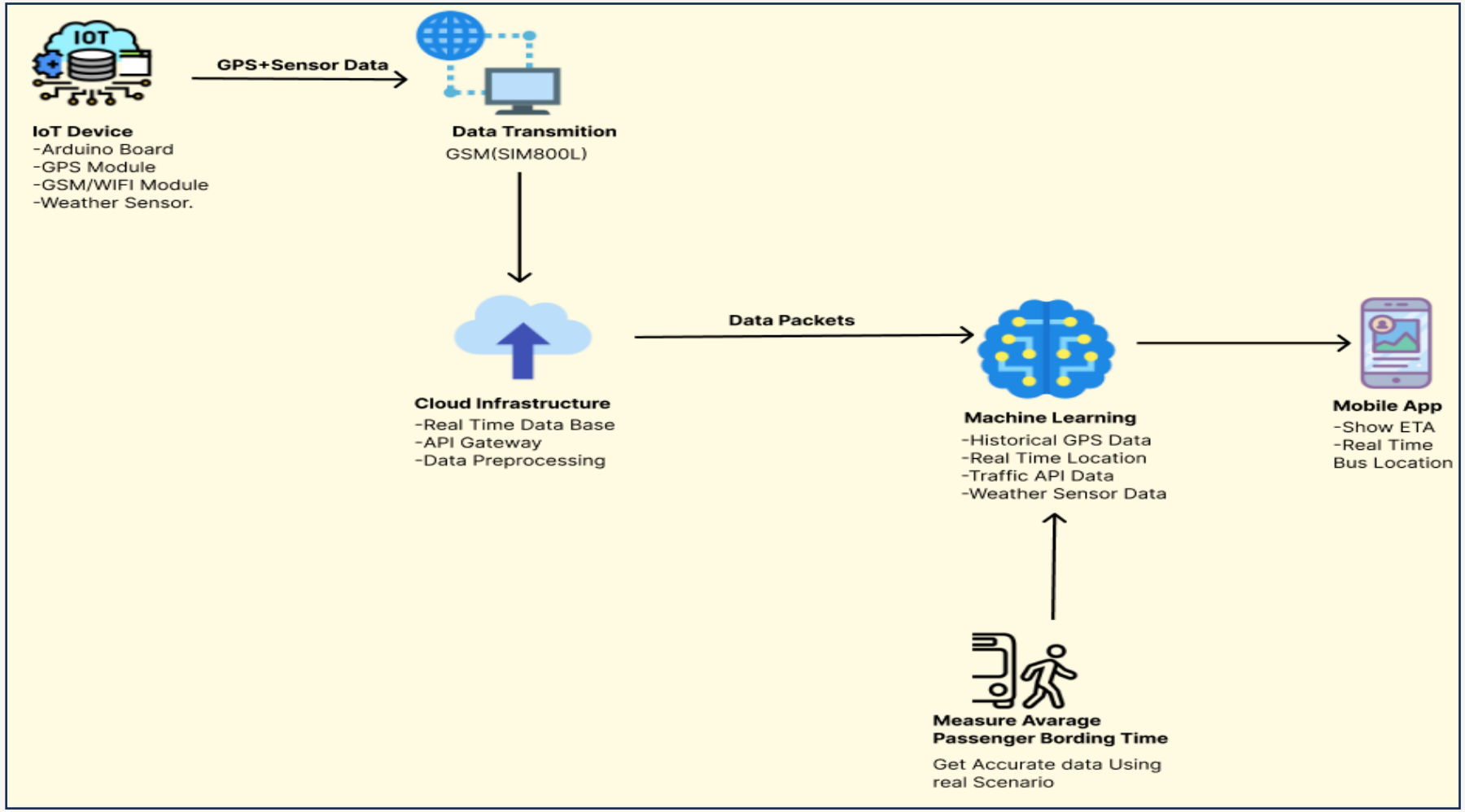
TensorFlow

Cloud Platforms

Firebase

Front-End

System Overview



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Central Fleet Management and Route Optimization Engine

Specialization: Information Technology

Project ID: 25-26J-267

Background

Shortcomings of Existing Systems:

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- Do not integrate **traffic, weather, or Passenger On boarding time**.
- Limited or no **multi-bus tracking**.
- Lack of affordable IoT-based implementations for developing regions.

Proposed Solution:

- IoT devices (GPS + GSM/Wi-Fi) provide **real-time data**.
- **LSTM model** predicts ETAs considering traffic, weather, and contextual factors.
- Supports **multi-bus tracking** with cloud integration.
- **Mobile/Web apps** deliver accurate arrival info to passengers and operators.

Research Problem

Sri Lanka bus transportation faces challenges in accurate ETA prediction, fare transparency, and driver safety monitoring. Passengers often experience delays without alternative options, face uncertain ticket prices, and have limited visibility into bus reliability or unsafe driving. There is a need for an integrated, real-time system that predicts arrival times, ensures transparent ticketing, and monitors driver behavior to improve passenger trust, safety, and overall service efficiency.

Research Gap

Features	[1]	[2]	[3]	[4]	Proposed System
EAT Prediction for passenger destination	✓	✗	✓	✗	✓
Transparent Ticket Pricing	✗	✓	✗	✗	✓
Driver Behavior Monitoring(IoT+ Alert)	✗	✗	✗	✗	✓
Real Time Driver Notification	✗	✗	✗	✓	✓
Ineractive Passenger Mobile UI	✗	✗	✗	✗	✓

Proposed Solution

An integrated smart public transport solution featuring real-time ETA prediction, transparent ticketing, and driver behavior monitoring. It uses IoT sensors, a passenger mobile/web platform, and automated driver notifications to enhance efficiency, safety, and convenience.

Tools & Technologies

TensorFlow

Python

node JS

Firebase

scikit learn

React

Requirements

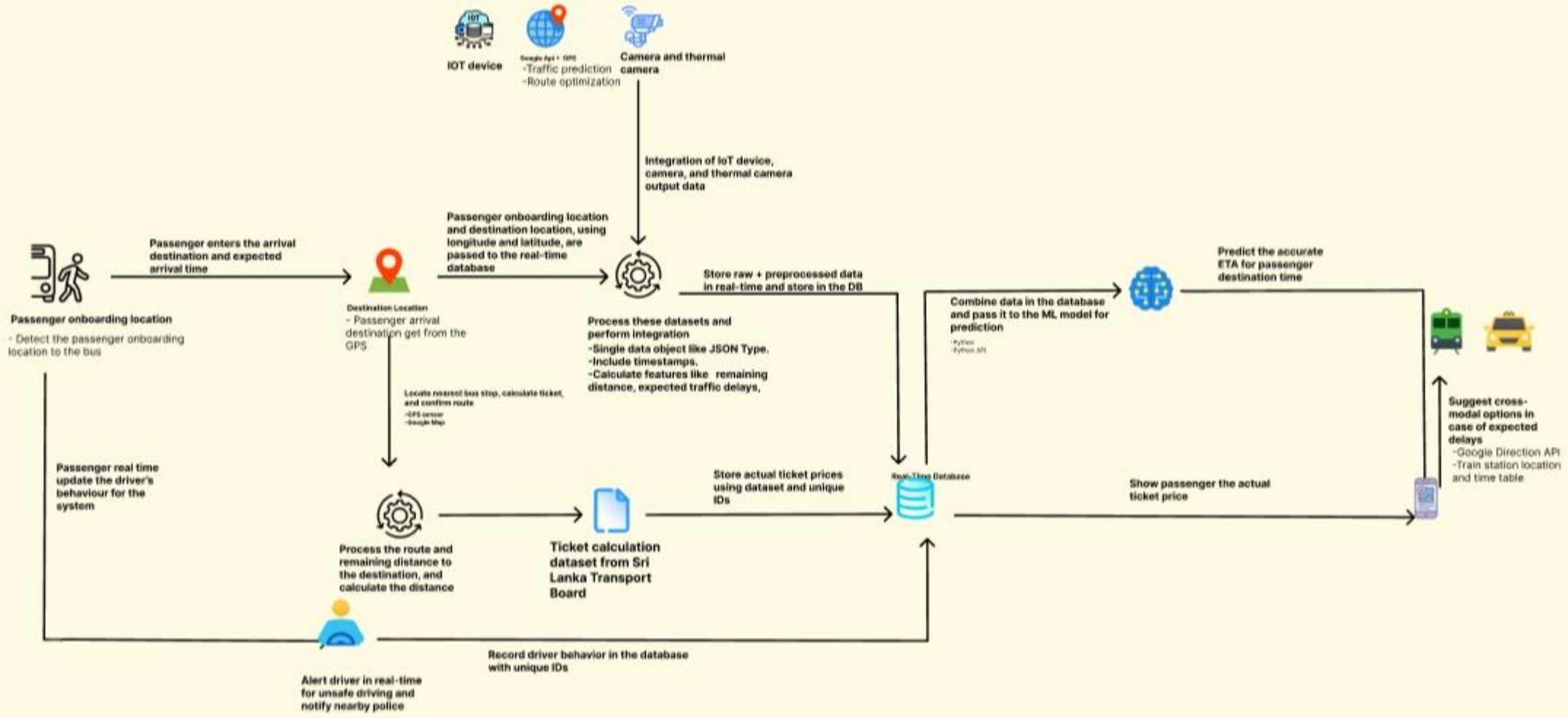
Functional Requirements

- Real-Time Data Collection
- Predictive ETA for passenger destination
- Cross-Modal Integration
- Fleet Operational Management
- Data Storage and Reporting
- User and System Security

Non Functional Requirements

- Performance
- Scalability
- Reliability and Availability
- Security
- Usability
- Maintainability
- Interoperability
- Efficiency

System Overview



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Mobile-Based Passenger Interaction System

Specialization: Information Technology

Project ID: 25-26J-267

Background

Public transport is vital for urban mobility, but most smart bus systems mainly focus on tracking and ETA, leaving key passenger needs unaddressed.

- Growing need for smart public transport solutions.
- Current systems focus mainly on tracking & ETA.
- No direct passenger → driver interaction.
- No crowd forecast for better journey planning.
- Need for a mobile-based passenger system.

Research Problem

Passengers often miss buses because there is no way to notify drivers in real-time. Current smart bus systems do not provide a mobile tool for requesting pickups or offer information about bus crowd levels, making journey planning difficult. Most systems focus on vehicles rather than the passenger experience.

Research Gap

Features	[1]	[2]	[3]	[4]	Proposed System
Pickup Request Functionality	✗	✗	✗	✗	✓
Real-Time Driver Notification	✓	✗	✗	✗	✓
Crowd-Aware Journey Suggestions (Date/Time Forecast)	✗	✓	✓	✗	✓
Past Data Forecasting (Passenger Load)	✗	✓	✓	✓	✓
IoT/Smart Transport Integration	✓	✓	✓	✓	✓
Interactive Web/Mobile Application	✗	✗	✗	✗	✓

Proposed Solution

- Mobile-based interaction system with pickup requests & crowd-aware journey suggestions

Tools & Technologies

Requirements

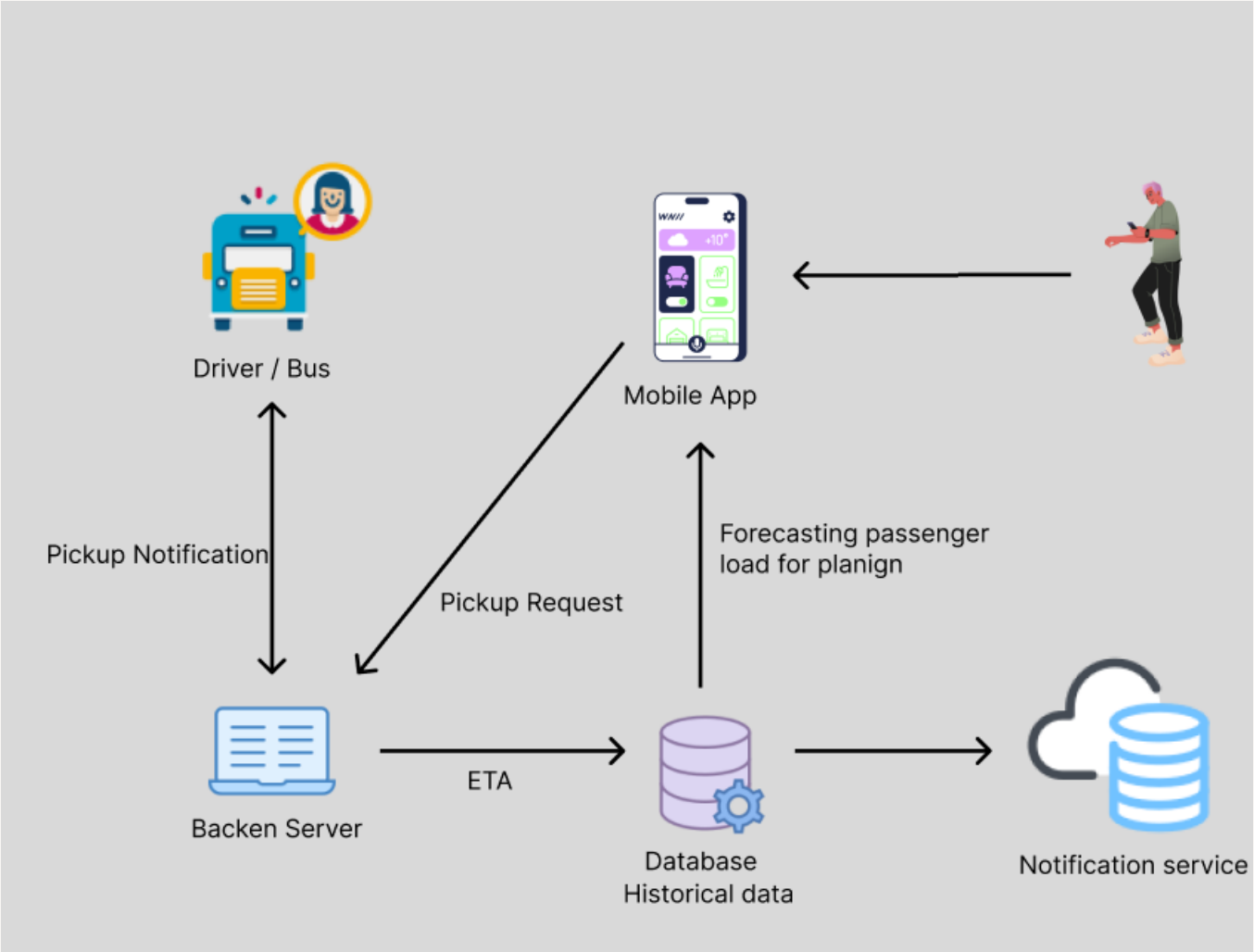
Functional Requirements

- Allow passengers to **send pickup requests** via a mobile app.
- Provide **crowd-aware journey suggestions** for passengers.
- Notify **drivers instantly** about passenger requests
- Store and manage data for **passenger requests, bus routes, and occupancy**.

NonFunctional Requirements

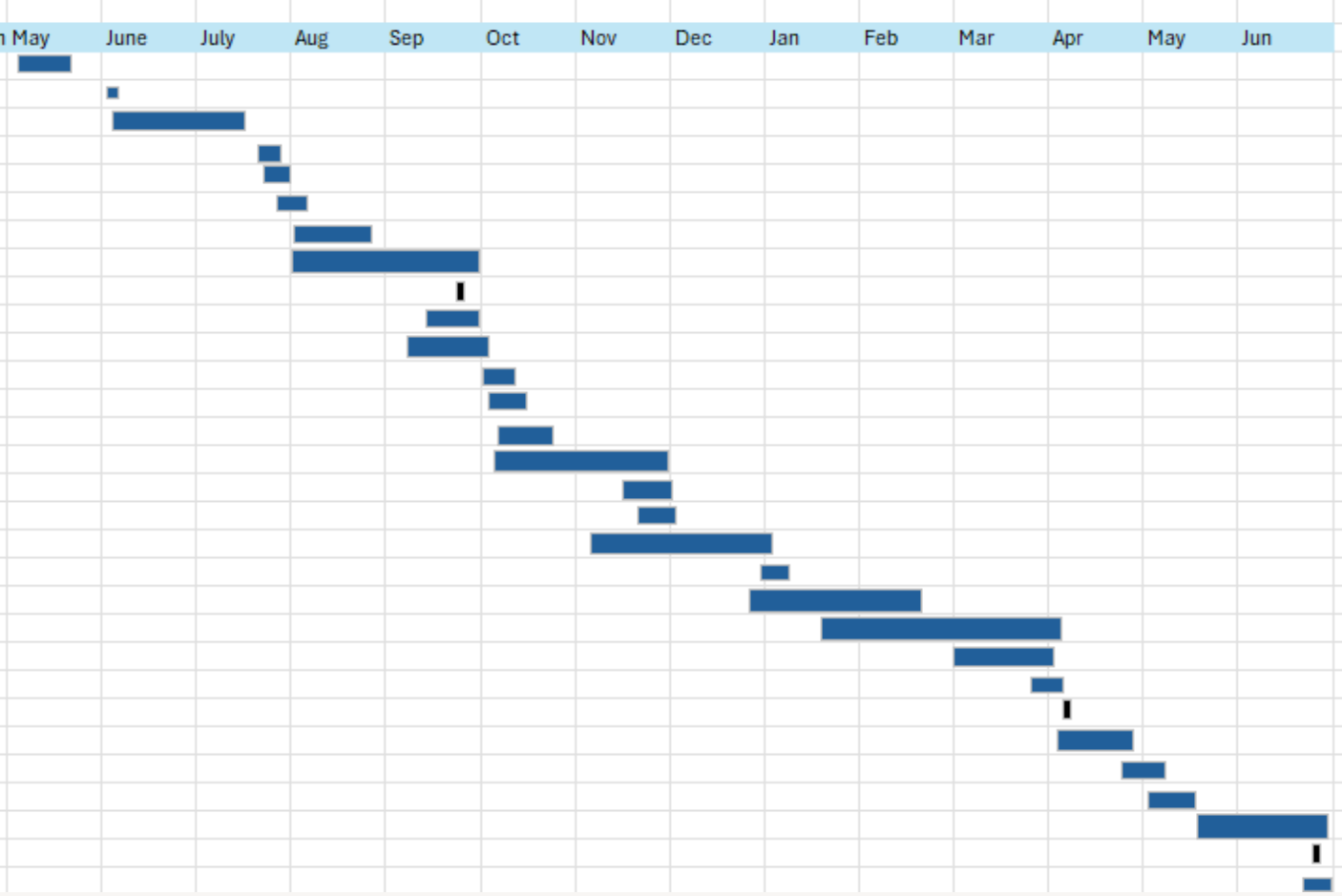
- **Real-time performance**: Updates must be reflected with minimal delay
- **Reliability**: System must handle multiple concurrent users without failure.
- **Scalability:** Should support increasing number of buses and passengers.
- **Security & Privacy:** Passenger data must be encrypted and anonymized.
- **Usability:** Mobile app should be intuitive and easy to use for all users.

System Overview

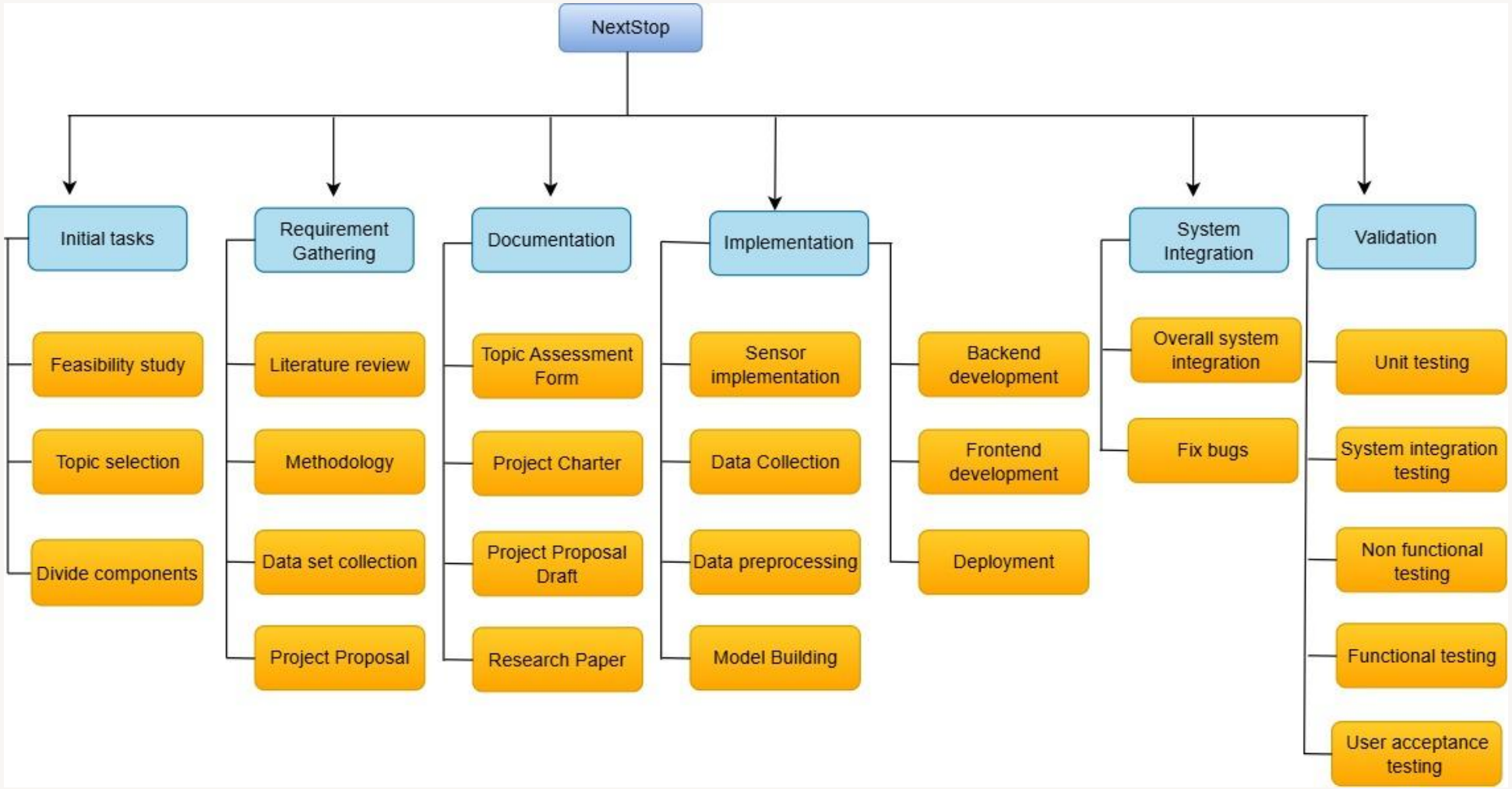


Project Time Line – Gantt Chart

ID	Name	Start Date	End Date	Duration
1	Topic selection	16 May, 2025	26 May, 2025	11 Days
2	Topic selection completed	26 May, 2025	27 June, 2025	2 Days
3	TAF Assessment	30 June, 2025	16 July, 2025	17 Days
5	Charter submission	17 July, 2025	23 July, 2025	7 Days
6	Feasibility study	24 July,2025	30 July, 2025	7 Days
7	Literature review	31 July, 2025	5 Aug, 2025	6 Days
8	Project proposal draft	6 Aug, 2025	15 Aug, 2025	10 Days
9	Proposal presentation	16 Aug, 2025	9 Sep,2025	25 Days
10	Proposal presentation complete	10 Sep, 2025	10 Sep, 2025	0 Days
11	Proposal report	11 Sep, 2025	19 Sep, 2025	9 Days
12	Finalizing dataset	20 Sep, 2025	30 Sep, 2025	11 Days
13	Finalizing methodologies	1 Oct, 2025	5 Oct, 2025	5 Days
14	Data preprocessing	6 Oct, 2025	10 Oct,2025	5 Days
15	Text extraction	11 Oct, 2025	17 Oct, 2025	7 Days
16	Model building / device building	18 Oct, 2025	8 Nov, 2025	22 Days
17	Model building completed	9 Nov,2025	16 Nov, 2025	8 Days
18	Progress presentation 1	17 Nov, 2025	19 Nov, 2025	5 Days
19	Model building 2	25 Nov, 2025	27 Dec, 2025	32 Days
20	Model building 2 completed	4 Jan, 2025	7 Jan, 2025	3 Days
21	Back End completion	8 Jan, 2025	6 Feb, 2025	30 Days
22	Frontend completion	7 Feb, 2025	10 Mar, 2025	32 Days
23	First integration	11 Mar, 2025	28 Mar, 2025	18 Days
24	Progress presentation 2	29 Mar, 2025	31 Mar, 2025	3 Days
25	Progress presentation 2 deliverd	1 Apr. 2025	1 Apr, 2025	0 Days
26	Integration copletion	2 Apr, 2025	28 Apr, 2025	27 Days
27	Unit testing	29 Apr, 2025	10 May, 2025	12 Days
28	Interation testing	11 May, 2025	25 May, 2025	15 Days
29	Final document	26 May, 2025	15 Jun, 2025	21 Days
30	Final document completed	16 Jun, 2025	16 Jun, 2025	0 Days
31	Final presentation and VIVA	20 Jun, 2025	25 Jun, 2025	6 Days



Work Breakdown Chart



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Vision based passenger counting & Occupancy

Specialization: Information Technology

Project ID: 25-26J-267

Background

- Urban public transport systems are vital for millions of commuters, but they often face challenges such as overcrowding, unpredictable arrival times, and lack of real-time seat availability information.
- These issues reduce passenger satisfaction and efficiency.
- Technology-based solutions (e.g., IoT, mobile apps) can help address these problems.

Research Problem

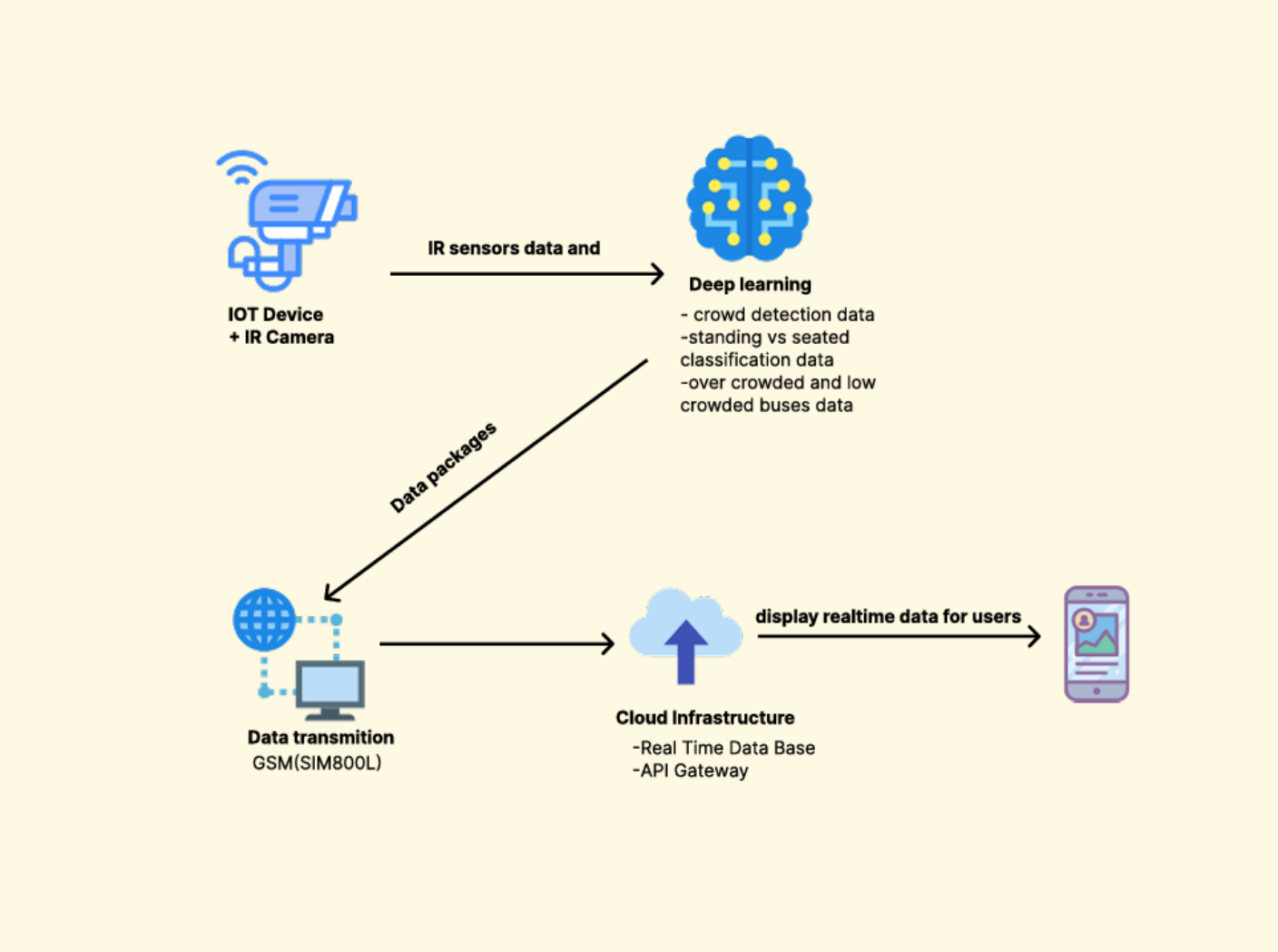
- Despite many studies on real-time crowd calculation in public transport, existing systems mainly rely on video-based solutions.
- There is currently no integrated system that combines thermal imaging and video cameras to deliver high-accuracy crowd estimation, real-time seat occupancy detection.
- No any system to route-specific bus crowd monitoring.

Research Gap

Vision based passenger counting & Occupancy

Features	[1]	[2]	[3]	[4]	Proposed System
Real-Time Crowd Calculation (Passenger Load)	✓	✗	Partial	✗	✓
IOT sensor integration	✓	✗	✓	Partial	✓
Crowd monitoring for authorize Parties.	✗	✗	✗	✗	✓
Interactive Mobile/Web Application	✗	✗	Partial	✗	✓
IoT/Smart Transport Integration	Partial	Partial	✗	Patial	✓

System diagram



Tools & Technologies

TensorFlow

Thermal cameras

Firebase

Google Cloud

Python

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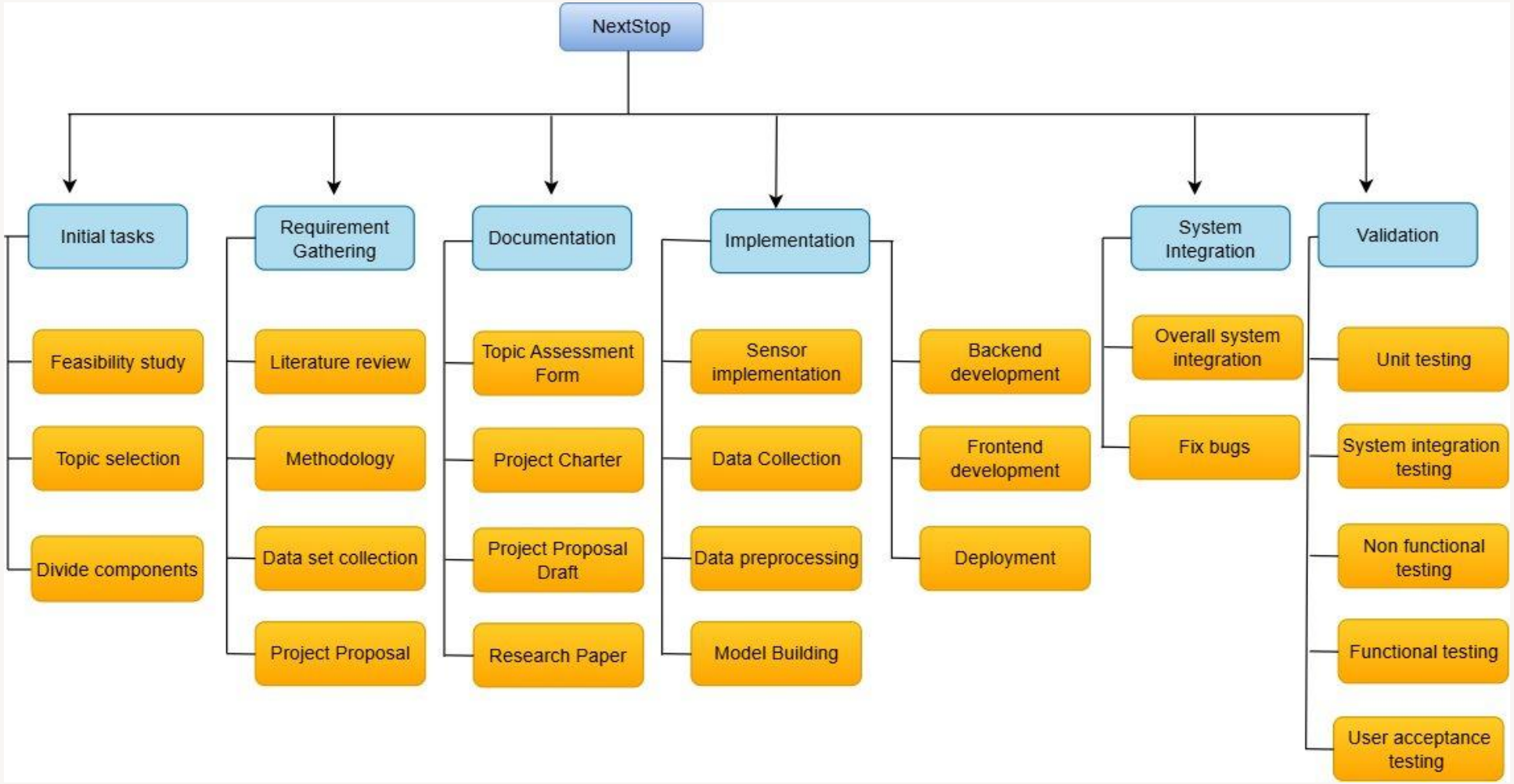
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